

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

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Program/Batch	B082-Networkwalks
Date	17 August 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3: Google Hacking Database Phase 4: Active Reconnaissance using Zenmap

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.

All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
Linux ip a	Local IP and MAC address identification
host	used to convert names to IP addresses and vice versa.
Crt.sh	Search tool used to lookup SSL/TLS from CT logs
theHarvester	open-source reconnaissance tool used to gather public information—such as emails, subdomains, and IP addresses
Zenmap	A GUI of Nmap tool
nmap	Used for active reconnaissance by scanning the devices, ports, versions, services and operating systems running on the devices

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: **host**, **WHOIS**, **WhatWeb**, **Nslookup**, **Curl**, **Wafw00f**, **curl.sh**, **netcraft** and **DNSRecon**. Each tool was used to collect a different type of information about the target.

```
[r_access@kali]~$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-09-15T12:38:54Z <<<
```

מחברת: ד"ר חגית גורן, מנהלת מחלקת המחקר, מרכז המחקר והמידע, מכון דוידסון לחינוך מדעי, אוניברסיטת תל אביב

Common Name	www.netcraft.com
Subject Alternative Names	Total SAN count: 1 www.netcraft.com
Certificate Trusted	Yes
Name Matches Domain	Yes
Validity	Jul 04, 2026 – Oct 02, 2026 (36 days from today)
Validity Period	OK (90 days)
OCSP/CRL Revocation	Good, not revoked (via CRL)
Issuer	Google Trust Services, US
Certificate Type	Domain Validation (Single Domain)
Algorithm / Key Type & Size	sha256 / EC 256 bits (secp256r1) Good signature Good key
Certificate Transparency	Compliant: 2 verified SCTs from 2 logs / 2 operators
Certificate Chain	Order: Valid Complete
Supported TLS Versions	TLSv1.2, TLSv1.3
DNS CAA	Yes – policy at netcraft.com <pre> issuetoall "pk1.goog; cnsignedtpechangelinesyes" flag: 0 issue "globaleidn.com" flag: 0 issue "amazon.com" flag: 0 issue "digicert.com; cnsignedtpechangelinesyes" flag: 0 issuetoall "amazon.com" flag: 0 issue "comodoca.com" flag: 0 issuetoall "digicert.com; cnsignedtpechangelinesyes" flag: 0 issue "pk1.goog" flag: 0 issue "letsencrypt.org" flag: 0 </pre>
Serial Number	00F5063887A0A1A7540E3F2B813C2B05F
SHA256 Fingerprint	0d577d3200739c3ebccbf03833937748b87e8f3440dc3cde2896a42d9
Revocation / Issuer URLs	CA issuers: http://r1.pk1.goog/wsl.crt CRL: http://r1.pk1.goog/wsl/crloq1-3p.crl

netcraft

Research Tools

Internet Intelligence

Platform Overview

REPORT FREQUENTLY

Share

Background

Site title	Not Available	Date first seen	January 2020		
Site rank	137206	Primary language	English		
Description	Not Present				

Network

Site	http://networktools.com/		Domain	networktools.com	
Network Owner	hostgator.com LLC		Nameserver	ns1.ns1.hostgator.com	
Hosting company	Hostford Digital		Nameserver organisation	godaddy.com	
Hosting country	 US		Nameserver organisation	www.enom.com	
IPv4 address	192.232.216.135 View/Feed 		Organisation	Unknown	
IPv4 autonomous systems	AS31898 		DNS admin	root@ns1.hostgator.com	
IPv6 address	Not Present		TCP Level Domain	Commercial entities (tag)	
IPv6 autonomous systems	Not Present		DNS Security Extensions	Enabled	
Reverse DNS	192.232.216-135.unifiedlayer.com				

IP delegation

IPv4 address (192.232.216.135)					
IP range	Country	Name	Description		
1 - 11111 - 8.8.8.8-16	 United States	IANA IPv4-MAPPED-ADDRESS	Internet Assigned Numbers Authority		
4 - 192.8.8.8-192.255.255.255	 United States	NET192	Various Registries (dominated by ARIN)		
4 - 192.232.192.8-192.232.255.255	 United States	HOBLOCK-8	HostGator.com LLC		
4 - 192.232.216.135	 United States	HOSTGATOR-192.232.216-135	HostGator.com LLC		

Then use the tools to interact with the DNS domains and translate the names and subdomains into ip addresses. For this use the tools **nslookup**, **host** and **dnsrecon**. This gives us a list of ip address to probe directly using active reconnaissance. Using **Nslookup**, I resolved the domain name to its IP address. The provided result identified **192.232.216.135**. Finally, I used **DNSRecon** to enumerate DNS records. The results provided information relating to name servers, mail servers, SPF/TXT records, service records and DNS software information.

```
(r_access@kali)-[~]
$ nslookup
> set type=mx
> networkwalks.com
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
networkwalks.com mail exchanger = 0 mail.networkwalks.com.

Authoritative answers can be found from:
>
```

```
(r_access@kali)-[~]
$ nslookup networkwalks.com
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
```

```
(r_access@kali)-[~]
$ host -l ns networkwalks.com
Using domain server:
Name: networkwalks.com
Address: 192.232.216.135#53
Aliases:

Host ns not found: 9(NOTAUTH)
; Transfer failed.
```

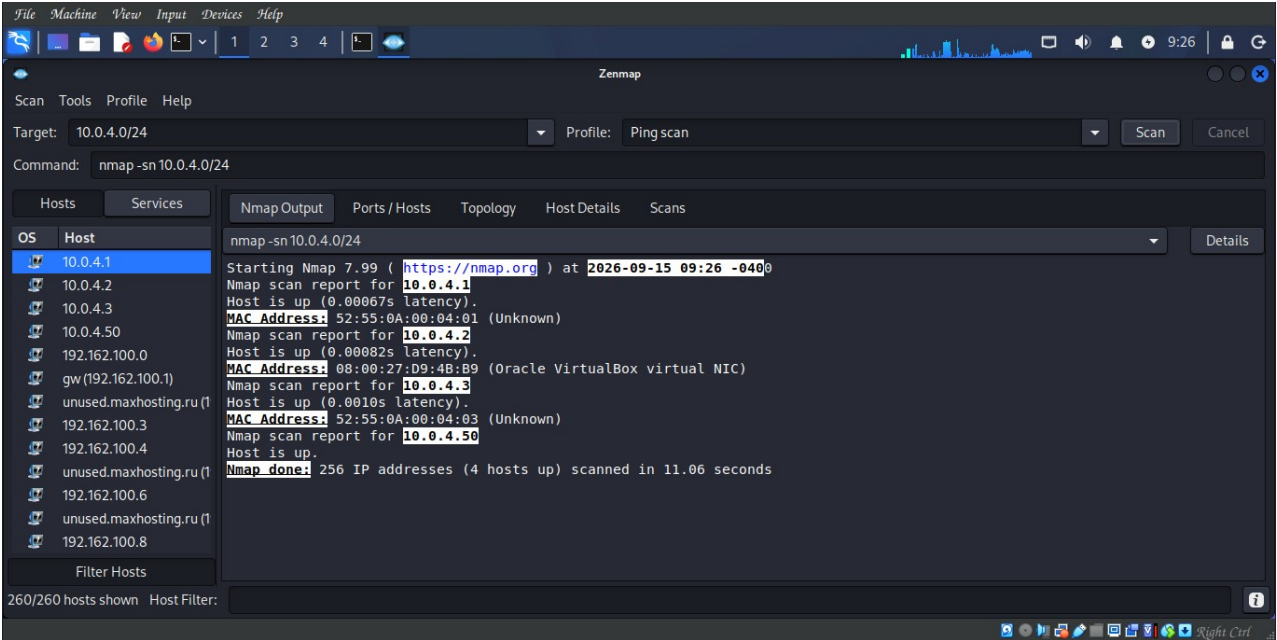
```
(r_access@kali)-[~]
$ dnsrecon -d networkwalks.com
2026-09-15T09:04:15.381842-0400 INFO Starting enumeration for domain: networkwalks.com
2026-09-15T09:04:15.382745-0400 INFO std: Performing General Enumeration against: networkwalks.com ...
2026-09-15T09:04:16.232662-0400 ERROR No answer for DNSSEC query for networkwalks.com
2026-09-15T09:04:16.966070-0400 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-09-15T09:04:17.863204-0400 INFO NS ns6136.hostgator.com 192.232.216.131
2026-09-15T09:04:18.544019-0400 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-09-15T09:04:18.544926-0400 INFO NS ns6135.hostgator.com 50.87.144.87
2026-09-15T09:04:19.135423-0400 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-09-15T09:04:20.194146-0400 INFO MX mail.networkwalks.com 192.232.216.135
2026-09-15T09:04:20.696306-0400 INFO A networkwalks.com 192.232.216.135
2026-09-15T09:04:21.753092-0400 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3Xemni
2026-09-15T09:04:21.755012-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:we
2026-09-15T09:04:22.646055-0400 INFO Enumerating SRV Records
2026-09-15T09:04:26.931163-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpa
2026-09-15T09:04:26.932713-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpa
2026-09-15T09:04:26.934586-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpa
2026-09-15T09:04:26.935568-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpa
2026-09-15T09:04:26.936477-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpa
2026-09-15T09:04:26.937452-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpa
2026-09-15T09:04:26.938560-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpa
2026-09-15T09:04:26.940007-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpa
2026-09-15T09:04:26.947235-0400 INFO 8 Records Found
2026-09-15T09:04:26.948621-0400 INFO Completed enumeration for domain: networkwalks.com
```

Now I have the exact ip addresses and web servers then used **WhatWeb** to identify technologies used by the website. The results identified **WordPress 7.0.4** and **WP Download Manager 3.3.58**, along with other information exposed by the website. Next, I used **Wafw00f** to determine whether a Web Application Firewall was protecting the website. The result identified **ModSecurity (SpiderLabs)**.

I first used the Linux `ip a` command to identify my local IP address and LAN subnet. I then entered the subnet into Zenmap and selected **Ping Scan** to identify active hosts.

The example results provided in the practical identified four live hosts:

- 10.0.4.1
- 10.0.4.2
- 10.0.4.3
- 10.0.4.50



The example results also included four MAC addresses.

After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend and saved the network topology in PDF format as required by the practical task.

4.3 Google Hacking Database

For the third activity, I practice the use of Google Hacking database which stores google dorks for various vulnerability search and penetration testing.

The first task is to use this database to find 10 vulnerable cameras along with their ip and google dork recorded:

IP Address	Relevant Dork	Username/Password(if any)
99.114.240.169	intitle:"Webcam" inurl:WebCam.htm	-
198.41.49.97	# Google Dork:	-

	intitle:"Device(IP CAMERA)" "language" -com net	
94.30.230.128	# Google Dork: intitle:"NoVus IP camera" -com	-
176.106.38.115	# Google Dork: intitle:"NoVus IP camera" -com	-
217.73.188.74	# Google Dork: intitle:"NoVus IP camera" -com	-
54.82.41.31	Google Dork: intitle:"Network Camera" inurl:main.cgi	-
109.233.191.130	# Google Dork: intitle:"webcamXP" inurl:8080	-
139.64.168.120	# Google Dork: intitle:"webcamXP" inurl:8080	-
72.199.200.5	# Google Dork: intitle:"webcamXP" inurl:8080	-
121.154.193.143	# Google Dork: intitle:"IP CAMERA" "User Login" "User Name" "Password" "Preview Stream"	-

The second task is to use the google dork to search mathematics ebook link with valid indexing:

Link	Dork
https://ebook-mecca.com/ebooks/Mathematics/?ND	intitle:"index of" (pdf epub mobi) "mathematics" -html -htm -php -asp
https://theswissbay.ch/pdf/Books/Mathematics/	intitle:"index of" (pdf epub mobi) "mathematics" -html -htm -php -asp
https://www.unm.edu/~megrad/Math/	intitle:"index of" (pdf epub mobi) "mathematics" -html -htm -php -asp

5. Risk Analysis

The activities performed during this assessment demonstrate how publicly available information and basic network discovery can be combined to build an initial picture of a target's attack surface. The reconnaissance phase identified domain registration information, DNS records, IP addresses, web technologies, security-related headers, and other publicly exposed information. The scanning phase demonstrated how active hosts within a local network can be identified and mapped. These findings do not by themselves confirm that the identified systems are vulnerable, but they provide information that could support further security testing or malicious reconnaissance.

5.1 Information Exposure Risk

The footprinting activities revealed several categories of information about the target domain, including name servers, DNS records, IP addresses, mail-related records, and technology information. DNS enumeration and certificate transparency searches can also reveal infrastructure and subdomains that may otherwise be overlooked during routine security monitoring.

From a security perspective, excessive information exposure can reduce the effort required by an attacker during the reconnaissance stage. Publicly available information should therefore be reviewed regularly to determine whether every exposed record, service, and technology identifier is necessary.

5.2 Web Technology Fingerprinting Risk

WhatWeb identified WordPress and the WP Download Manager plugin on the target website. Technology fingerprinting does not indicate a vulnerability on its own, but publicly identifiable software and plugin versions can help an attacker determine which technologies are present and research whether known weaknesses exist.

The presence of ModSecurity was also identified through Wafw00f. A Web Application Firewall provides an additional security layer, but its presence should not be considered proof that the web application itself is secure.

5.3 Web Application Information Exposure

HTTP header inspection using Curl identified the WordPress REST API endpoint `/wp-json/`. The availability of such an endpoint is normal for WordPress installations and is not inherently a vulnerability. However, exposed application interfaces should be reviewed to ensure that they disclose only the information and functionality intended to be publicly accessible.

5.4 Network Discovery Risk

The Zenmap activity demonstrated that active hosts and their IP and MAC addresses can be identified within a local subnet. In a real organizational environment, unnecessary exposure of network devices and services could provide an attacker with useful information for subsequent attack stages.

Network discovery should therefore be considered part of an organization's attack-surface management process. Devices, services, and network segments should be documented and monitored so that unexpected systems can be identified quickly.

5.5 Overall Risk Assessment

The assessment primarily identified **reconnaissance and information-exposure risks rather than confirmed exploitable vulnerabilities**. The findings show that an attacker can collect meaningful information about publicly accessible infrastructure and use active scanning to identify systems on an accessible network.

The overall risk therefore depends on how this information is combined with other weaknesses, such as outdated software, insecure configurations, weak authentication, exposed services, or inadequate network segmentation. Further vulnerability assessment would be required before assigning a definitive severity to individual technical weaknesses.

6. Recommendations

Based on the reconnaissance and scanning activities performed, the following security improvements are recommended:

6.1 Minimize Unnecessary Information Exposure

Regularly review publicly available DNS records, subdomains, certificates, server information, and other infrastructure details. Remove or restrict records that are no longer required and avoid unnecessarily exposing internal infrastructure information.

6.2 Keep Web Technologies Updated

Maintain WordPress, plugins, themes, web servers, and supporting software at supported and security-patched versions. In particular, software identified during technology fingerprinting should be regularly checked against vendor security advisories and updated when appropriate.

6.3 Review WordPress REST API Exposure

Review the /wp-json/ endpoint and determine what information is publicly accessible through the WordPress REST API. Public functionality should remain available where required, while sensitive information and administrative functionality should require appropriate authentication and authorization.

6.4 Properly Configure the Web Application Firewall

Continue using the identified ModSecurity protection, but regularly review its configuration, rules, logging, and effectiveness. The WAF should complement secure application development and server hardening rather than serve as the only defensive control.

6.5 Perform Regular External Attack-Surface Reviews

Conduct periodic authorized reconnaissance to identify newly exposed IP addresses, subdomains, services, certificates, and technologies. This can help identify changes introduced during system deployments or configuration updates.

6.6 Strengthen Network Segmentation

Internal networks should be appropriately segmented so that discovering one active host does not automatically provide broad access to other systems. Critical devices and services should be placed in suitable network segments with access controlled through firewalls and ACLs.

6.7 Monitor Unknown or Unexpected Devices

Organizations should maintain an inventory of authorized network devices and periodically compare it with network-discovery results. Unknown hosts should be investigated rather than being treated as trusted devices.

6.8 Conduct Further Vulnerability Assessment

The activities in this report focused primarily on reconnaissance, footprinting, and network discovery. A subsequent authorized assessment should evaluate identified services and applications for actual vulnerabilities, insecure configurations, authentication weaknesses, and outdated components. Any further testing should remain within the agreed scope and authorization.

7. Conclusion

This assessment demonstrated the progression from passive reconnaissance and footprinting to active network discovery. The footprinting phase showed how tools such as WHOIS, WhatWeb,

Nslookup, DNSRecon, Wafw00f, Curl, crt.sh, Netcraft, and theHarvester can be used to build an initial picture of a target's publicly visible infrastructure. The scanning phase then demonstrated how Zenmap can identify active hosts and assist in visualizing a local network topology.

The assessment also demonstrated that reconnaissance can reveal useful information without directly exploiting a system. Domain information, DNS records, IP addresses, web technologies, security controls, and network hosts can provide an attacker with valuable knowledge for planning later stages of an attack.

However, the findings in this report should not be interpreted as confirmed vulnerabilities. The activities primarily established the available attack surface and identified information that should be reviewed from a security perspective. No exploitation was performed as part of the documented activities.

Overall, the exercise highlights the importance of minimizing unnecessary information exposure, maintaining updated software, reviewing publicly accessible services, monitoring network assets, and performing regular authorized security assessments. These controls can reduce the amount of information available to unauthorized parties and make subsequent attack stages more difficult.

Author

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